

# Flow Measurement: Pipe-Flow Meters and Free-Jet Velocity Profiles

Calibrating differential-pressure flowmeters and mapping a turbulent air jet with a Pitot tube

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|--------|-----------------------------------------------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------|
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| Course | ME 343-004 Mechanical Lab I, NJIT                                           | Instructor | Prof. B. S. Mani · TA: Saba Mirahsani                                                                                                    |
| Team   | Team 4: Luis Ramirez, Samuel Cohen, Edwin Ortezt, Abhin Ghandi, Ramid Brown | Equipment  | Pipe-flow bench (rotameter, venturi, orifice, turbine meter, manometer bank, weigh tank), fan-driven jet, Pitot tube, inclined manometer |

## PROJECT SUMMARY

This experiment measured internal pipe flow and a free air jet using several flow-measurement methods. On the pipe-flow bench, weigh-tank reference flows were used to calibrate the rotameter and to evaluate the venturi and orifice meters against Bernoulli’s equation. The orifice discharge coefficient rose with flow rate and leveled off near 0.60–0.61, matching the standard value for a sharp-edged orifice. A Pitot tube and inclined manometer mapped the velocity profile of a fan-driven jet 100, 250, and 400 mm downstream: the centerline velocity dropped from 37.9 to 28.6 m/s as the jet spread, and the normalized profiles collapsed onto a similar shape, demonstrating the self-similarity of turbulent jets.

## 1. Introduction

Precise flow measurement is essential for managing pipelines, testing aerodynamics, and controlling processes. This experiment connected fluid mechanics theory to real instruments by studying two flow cases — internal pipe flow and a free air jet — and comparing each instrument’s readings with predictions based on Bernoulli’s equation to understand measurement accuracy, discharge losses, and turbulent jet behavior.

## 2. Theory & Instruments

Table 1. Flow measurement instruments

| Instrument    | Principle                                                                                                                                             |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Venturi meter | A gradual contraction accelerates the flow; the pressure drop between inlet and throat gives the flow rate. Losses are small, so $C_d$ is close to 1. |
| Orifice meter | An abrupt plate constriction creates a larger pressure drop and more turbulent loss, giving a lower $C_d$ (about 0.6).                                |
| Rotameter     | A float rises in a tapered tube until drag balances its submerged weight; float height indicates flow rate.                                           |
| Turbine meter | Rotor speed is proportional to fluid velocity; needs calibration for friction and flow irregularities.                                                |
| Pitot tube    | Measures stagnation minus static pressure (dynamic pressure), giving local velocity from Bernoulli’s equation.                                        |

For a differential-pressure meter, the ideal flow rate is  $Q_{ideal} = A_2 \sqrt{[2\Delta p / (\rho(1 - (A_2/A_1)^2))]}$ , and the discharge coefficient  $C_d = Q_{actual}/Q_{ideal}$  accounts for real losses. For the Pitot tube, local velocity is  $u = \sqrt{(2\Delta p/\rho_{air})}$ , where

$\Delta p$  is found from the manometer height difference.

### 3. Procedure

On the pipe-flow bench, flow was set at ten rates, and for each one the actual flow was found from the weigh tank while manometer readings were taken across the venturi and orifice meters and the rotameter float height was recorded. For the jet, the Pitot tube was traversed laterally across the jet at 100, 250, and 400 mm downstream of the nozzle, recording the inclined manometer at six positions.



Figure 1. Pipe-flow bench and manometer bank



Figure 2. Rotameter



Figure 3. Pitot tube in the jet

### 4. Results

#### Orifice meter calibration

Table 2. Orifice meter ( $d_1 = 51.9$  mm,  $d_2 = 20$  mm)

| Q actual (L/min) | 5.0   | 6.4   | 7.9   | 9.4   | 10.4  | 11.7  | 13.4  | 14.6  | 16.0  | 17.5  |
|------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Q ideal (L/min)  | 15.79 | 15.33 | 17.30 | 18.49 | 19.80 | 21.19 | 22.65 | 24.17 | 26.69 | 28.50 |
| $C_d$            | 0.32  | 0.42  | 0.46  | 0.51  | 0.53  | 0.55  | 0.59  | 0.60  | 0.60  | 0.61  |

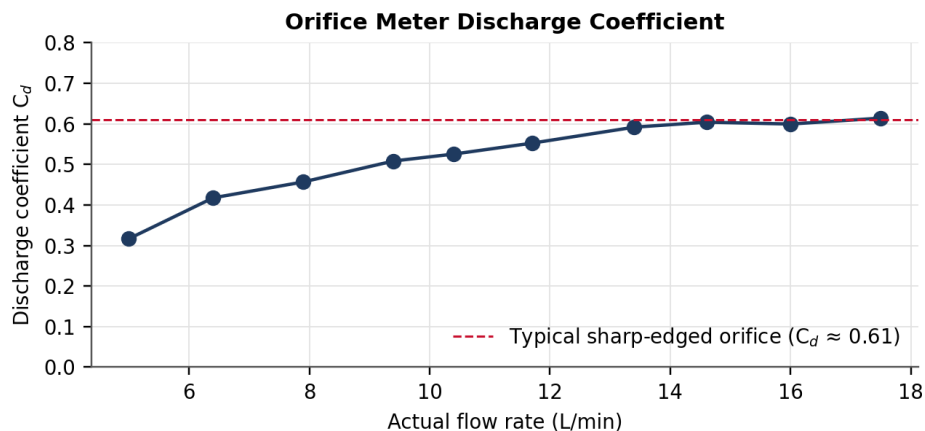


Figure 4. Orifice discharge coefficient vs. actual flow rate.

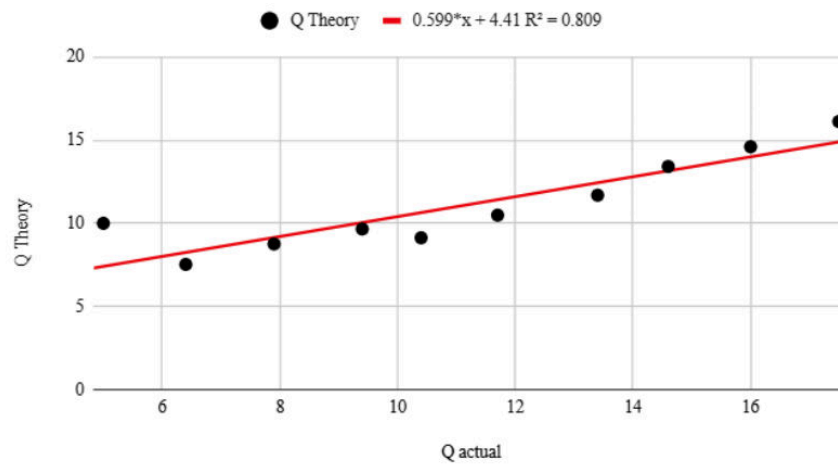


Figure 5. Venturi meter: theoretical vs. actual flow rate (team chart).

## Free-jet velocity profiles

Table 3. Pitot tube velocities across the jet (y/b uses b = 60 mm)

| Lateral position y (mm) | 1     | 20    | 26    | 35    | 45    | 60    |
|-------------------------|-------|-------|-------|-------|-------|-------|
| u at 100 mm (m/s)       | 37.89 | 36.86 | 34.71 | 23.75 | 19.61 | 12.40 |
| u at 250 mm (m/s)       | 35.44 | 34.34 | 30.38 | 26.79 | 19.61 | 11.32 |
| u at 400 mm (m/s)       | 28.64 | 26.31 | 24.80 | 20.25 | 17.54 | 11.32 |

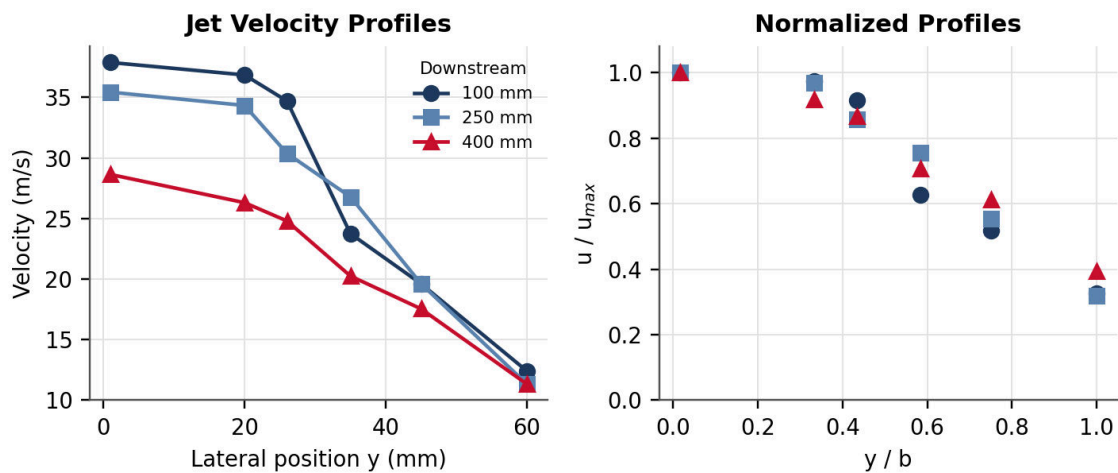


Figure 6. Jet velocity profiles at three downstream stations (left) and the same data normalized by centerline velocity (right).

## 5. Discussion

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**Pipe-flow meters.** Pressure differences across both meters increased steadily with flow rate, as Bernoulli's equation predicts. The orifice discharge coefficient climbed from 0.32 at the lowest flow to a plateau of 0.60–0.61 above about 13 L/min, which is the textbook value for a sharp-edged orifice. The low values at low flow come from the very small manometer differences there, where a reading error of a millimeter or two is a large fraction of the total. The venturi data followed a linear trend but with more scatter ( $R^2 = 0.81$ ). The rotameter showed a strong linear relationship between float height and weigh-tank flow rate.

**Free jet.** The centerline velocity decreased with distance from the nozzle, from 37.9 m/s at 100 mm to 35.4 m/s at 250 mm and 28.6 m/s at 400 mm. As the jet travels, it entrains the surrounding still air and spreads laterally, so its momentum is shared by more fluid and the core slows down. Velocity also dropped sharply toward the jet edge, marking the shear layer between the fast core and the surrounding air. When each profile was divided by its own centerline velocity, the three stations fell onto a similar curve, illustrating the self-similarity of turbulent jets.

## 6. Conclusion

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The experiment validated several flow measurement techniques for both internal pipe flow and a free air jet. The rotameter calibrated linearly against the weigh tank, the venturi and orifice meters followed Bernoulli trends, and the orifice discharge coefficient settled at the expected value of about 0.61. The Pitot tube survey showed the jet slowing and spreading downstream while keeping a similar normalized profile. With careful calibration, the experimental results closely matched analytical predictions, bridging fluid mechanics theory and practical measurement.

## References

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- Holman, J. P. (2012). *Experimental methods for engineers* (8th ed.). McGraw-Hill.
- Zhu, C. (2009). *ME 343 laboratory introduction*. New Jersey Institute of Technology.